

DECISION PROCEDURES FOR SOME FRAGMENTS OF SET THEORY

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1. INTRODUCTION.

The present note describes several decision techniques applicable to elementary set-theoretic statements of forms likely to occur within extended proof-verifications formulated in any language which allows use of set theoretic constructs. (A proof verifier for such a language is currently being developed at our university). The algorithms to be described can determine the validity of arbitrary collections of statements involving the propositional connectives, the elementary set-boolean operations intersection, union, difference, etc., the set comparisons $a = b$ and $a \subseteq b$, the membership relation $a \in b$, the singleton-former $\{a\}$, and the set cardinality operator $\#a$, together with integer addition, subtraction, and comparison. Two algorithms are presented; the first, which has been implemented in the programming language SETL with some useful optimizations, handles only the restricted case in which neither the singleton-former nor the cardinality operator appears, but is more efficient than the second algorithm. In the last section it is shown that the algorithms presented can be extended to handle statements involving map operators such as domain, range, and the inverse operator f^{-1} together with the special map predicates *singlevalued*(f) and *one-one*(f). See Pratt [7] for a statement emphasizing the importance of built-in knowledge of important special cases to the design of broad-spectrum proof verifiers.

2. SYNTAX AND SEMANTICS OF MULTILEVEL SYLLOGISTIC.

The symbols of the language of multilevel syllogistic are:

- (1) The parentheses $)$ and $($;
- (2) a denumerable sequence of *variables* $x_1, x_2, \dots$;
- (3) the *constant* $\emptyset$;
- (4) the *binary operators* $\cup, \cap, -$;
- (5) the *binary relators* $=, \subseteq, \in$;
- (6) the usual logical *connectives* $\neg, \&, \vee, \rightarrow, \leftrightarrow$.

In what follows, $*$ will indicate any of the binary operators.

The set T of *terms* is the smallest superset of $\{\emptyset, x_1, x_2, \dots\}$ such that if t_1 and t_2 are in T then $(t_1 * t_2)$ is also in T .

The *atomic formulae* are the expressions of the form

$$(t_1 = t_2), (t_1 \subseteq t_2), (t_1 \in t_2),$$

where t_1 and t_2 are terms.

Negated and unnegated atomic formulae are called *literals*.

The set of *formulae* is the smallest set containing all the atomic formulae and closed with respect to the logical connectives. $t_1 \neq t_2$ and $t_1 \not\subseteq t_2$ abbreviate $\neg(t_1 = t_2)$ and $\neg(t_1 \subseteq t_2)$ respectively.

Example of a formula:

$$(x \cup z \not\subseteq (\emptyset - (z - x))) \ \& \ (z \in y \cup x) \rightarrow (x \not\subseteq \emptyset) \vee (x \cap z \neq y) \vee (y \subseteq \emptyset).$$

An *interpretation* M is an assignment of a set to each variable. Once these sets are assigned, constant and operator signs of our language are interpreted in the usual set-theoretic sense, namely $\emptyset, \cup, \cap, -, \subseteq, \in$, designate the empty set, union, intersection, set difference, inclusion, and membership respectively.

With these rules every interpretation assigns a unique set value to each term and a unique truth value to each formula. M is said to be a *model* for each formula which is true in M . If every (some) interpretation is a model for the formula p , then p is said to be *valid* (*satisfiable*).

If $p * q$ is valid then p and q are said to be (logically) *equivalent*.

If $p \rightarrow q$ is valid then q is said to be a (logical) *consequence* of p .

M is called a *singleton model* if every set variable is interpreted as a subset of $\{\emptyset\}$.

3. A DECISION PROCEDURE FOR MULTILEVEL SYLLOGISTIC.

Our problem is to check a formula p for validity. It can easily be shown that this is equivalent to checking satisfiability of a conjunction q of literals having one of the following forms:

- (*) $x = y * z$
- ($\neq$) $x \neq y$
- ($\in$) $x \in y, \quad x \not\subseteq y$

where x, y, z are variables.

Let q^* be the conjunction of all the literals in q having the form $(*)$ and let $V_{\in}$ be the set of all the variables x in q which appear as left-hand sides of literals of the form $(\in)$. For each x in $V_{\in}$ consider the conjunction L_x of all the literals in q of the form $(\in)$ whose left-hand side is x . Let L'_x be obtained from L_x by replacing $x \in y$ and $x \notin y$ by $y \neq \emptyset$ and $y = \emptyset$ respectively. The following algorithm (see [3]) decides the satisfiability of q :

- STEP 1. For every pair of distinct variables x, y in q , determine if the equality $x = y$ is a logical consequence of q^* , by testing $q^* \ \& \ x \neq y$ for the existence of a singleton model. Put x and y into the same equivalence class whenever q^* entails $x = y$. Choose a representative variable z from each class and replace all the other variables in the class by z in q .
- STEP 2. If any of the literals in q has the form $x \neq x$ then return 'UNSATISFIABLE'.
- STEP 3. If $V_{\in}$ is empty then return 'SATISFIABLE'.
- STEP 4. If, for each x in $V_{\in}$, we can find a singleton model M_x for $q^* \ \& \ L'_x$ in such a way that the relationship $x < y$ defined by $x < y$ iff $M_x \ y \neq \emptyset$ extends to a total ordering on all the variables of q , then return 'SATISFIABLE'; otherwise return 'UNSATISFIABLE'.

Example: Let p be $(x \in y \cup z) \ \& \ (y \in w) \ \& \ (z \in w) \rightarrow w \notin x$. Then $\neg p$ is $(x \in y \cup z) \ \& \ (y \in w) \ \& \ (z \in w) \ \& \ (w \in x)$ and q is

$$u = y \cup z \ \& \ x \in u \ \& \ y \in w \ \& \ z \in w \ \& \ w \in x.$$

The only singleton models for $q^* \ \& \ L'_x$ are

Case (1): $u \neq \emptyset, y \neq \emptyset, z$ arbitrary; Case (2): $u \neq \emptyset, z \neq \emptyset, y$ arbitrary.

In case (1) $x < u, x < y$ and in case (2), $x < u, x < z$. In both cases $w < x, y < w$ and $z < w$. Thus in the first case we have a cycle $x < y < w < x$ whereas in the second case we have a cycle $x < z < w < x$. Hence we can conclude that q is unsatisfiable, i.e. that p is valid.

Remark 1. It is important to note that the set of statements $q^* \ \& \ L'_x$ appearing in Step 4 has a singleton model iff the result of

substituting

$x \leftrightarrow y$	for $x = y$
$x \vee y$	$x \cup y$
$x \& y$	$x \cap y$
$x \& \neg y$	$x - y$
$\neg x$	$x = \emptyset$

in $q^* \& L'_x$ has a propositional model (the same assertion holds for the set of statements $q^* \& x \neq y$ mentioned in step 1). Thus if a singleton model for $q^* \& L'_x$ (or $q^* \& x \neq y$) exists it can be found by any standard decision procedure for propositional calculus [2].

Remark 2. The convexity (in Oppen's sense, see [6]) of the theory we consider is a consequence of the following easily proved assertion:

If q is satisfiable and $x_1 = y_1 \vee x_2 = y_2 \vee \dots \vee x_n = y_n$ is a logical consequence of q then for some $i = 1, 2, \dots, n$, $x_i = y_i$ is a logical consequence of q^* .

Indeed, if q is satisfiable whereas $q \& x_1 \neq y_1 \& \dots \& x_n \neq y_n$ is unsatisfiable, then since this unsatisfiability will be necessarily detected at Step 2, it must be caused by some literal $x_i \neq y_i$ for which $x_i = y_i$ is a logical consequence of q^* .

4. AN EXTENDED MULTILEVEL SYLLOGISTIC.

In this section we describe an algorithm for deciding formulae of an extended multilevel syllogistic which includes the singleton-former and cardinality operator $\#x$ together with arithmetic addition, subtraction, and comparison in addition to the operators considered previously. In this extended language set terms are built as described in section 2, but we have an additional set construction, namely

$\{t\}$ is a set term (designating a singleton set) whenever t is a set term. Integer terms are built up using the binary operators $+$ and $-$ from a new infinity of integer variables, the constants 0, 1, and terms of the form $\#t$, where t is a set term. We also allow two new kinds of atomic formulae $j \leq k$ and $j = k$, where j and k are integer terms.

The validity problem for the new language reduces to that of checking satisfiability of a conjunction of literals of the following forms:

(*)	$x = y * z$
($\neq$)	$x \neq y$
($\{ \}$)	$y = \{x\}$

- (#) $j = \#x$
 (A) $0 \leq j, j = k-l, j = 1, j = 0$

where x, y, z are set variables and j, k, l are integer variables.

The following theorem (see [3]) defines our decision algorithm in rough outline:

THEOREM 1. Let q be a conjunction of literals having one of the above forms. Let V be the collection of all the set variables in q and let P be the collection of all nonempty subsets S of V such that the interpretation

$$M_x = \begin{cases} \emptyset & \text{if } x \notin S \\ \{\emptyset\} & \text{otherwise} \end{cases}$$

is a singleton model for q^* . Associate an integer variable v_S with each S in P and put $\Delta(a, b) = \sum_{\substack{a \in S \\ b \notin S}} v_S + \sum_{\substack{a \notin S \\ b \in S}} v_S$.

Then q is satisfiable if and only if there is a partial ordering $<$ on P such that the following system of arithmetic conditions has a solution in integers such that

if $(y = \{x\})$ is in q , $x \in S$, $y \in S'$, and $S \not\prec S'$ then
 either $v_S = 0$ or $v_{S'} = 0$:

- (1) take all the conditions of the form (A) in q ;
- (2) add the conditions $\Delta(x, y) > 0$, for all $(x \neq y)$ in q ;
- (3) add the conditions $\Delta(x, \emptyset) = j$, for all $(j = \#x)$ in q ;
- (4a) add the conditions $\Delta(y, \emptyset) = 1$, for all $(y = \{x\})$ in q ;
- (4b) add the conditions $\Delta(x, x') = 0 \wedge \Delta(y, y') = 0$,
 for all $(y = \{x\}), (y' = \{x'\})$ in q ;
- (5) finally add the conditions $v_S \geq 0$, for all S in P .

It follows from the decidability of Presburger arithmetic [8] that the satisfiability of these conditions is mechanically verifiable. In view of the decidability of the additive theory of cardinals [13], this remains true if we allow arbitrary cardinal numbers to appear in place of integers. For application of the decision method of this section it is worth noting that finite set formers and ordered tuples can be expressed using the singleton-former, as is shown by the following rather standard recursive definitions:

- (a) $\{ \} = \emptyset$
 $\{e_1, e_2, \dots, e_n\} = \{e_1\} \cup \{e_2, e_3, \dots, e_n\}$;
- (b) $\langle e \rangle = e$
 $\langle e_1, e_2, \dots, e_n \rangle = \{\{e_1\}, \{e_1, \langle e_2, e_3, \dots, e_n \rangle\}\}$.

5. A FURTHER EXTENSION OF MULTILEVEL SYLLOGISTIC.

We can also extend multilevel syllogistic by introducing maps as a new sort of variable, together with domain, range, and restricted range operators which take maps into sets. For this extension, we introduce a new infinity of map variables $f, g, \dots$, and two new map operators D and R . Set terms can be built as before but we have the additional set terms Df , Rf , and $f[t]$ where f is any map variable and t is a set term (here Df and Rf respectively denote the domain and the range of f ; $f[t]$ denotes the range of the restriction of f to t).

The validity problem for the language thus defined reduces to that of checking satisfiability of a conjunction q of literals of the following forms:

- (*) $x = y * z$
- ($\neq$) $x \neq y$
- ($\in$) $x \in y, \quad x \notin y$
- ($[]$) $x_{f_y} = f[y]$
- (D, R) $x_{Df} = Df, \quad x_{Rf} = Rf,$

where $x, y, z, x_{f_y}, x_{Df}, x_{Rf}$ are set variables.

Without loss of generality we can suppose that f occurs in q iff literals involving f of each of the three types ($[]$), and (D, R) occur in q . For every f in q let $y_1, y_2, \dots, y_k$ be all the variables appearing on the right-hand side of literals of the form $x = f[y]$ in q , and let y^+ and y^- stand for $y \cap x_{Df}$ and $x_{Df} - y$ respectively. Also for each such f introduce new set variables $s_{i_1 i_2 \dots i_k}$, where each i_j takes one of the two values $+, -$, representing $f[y_1^{i_1} \cap y_2^{i_2} \cap \dots \cap y_k^{i_k}]$, and add the following formulae to q ;

$$(1) \quad y_1^{i_1} \cap y_2^{i_2} \cap \dots \cap y_k^{i_k} = \emptyset \leftrightarrow s_{i_1 i_2 \dots i_k} = \emptyset;$$

$$(2) \quad x_{f_{y_j}} = \bigcup_{i_j = +} s_{i_1 i_2 \dots i_k};$$

$$(3) \quad x_{Rf} = \bigcup_{i_1 i_2 \dots i_k} s_{i_1 i_2 \dots i_k};$$

Delete all the literals of the forms ($[]$) and (D, R). Let $\bar{q}$ be the resulting formula.

THEOREM 2. q is satisfiable iff $\bar{q}$ is satisfiable.

Proof: If q has a model M , then M can be extended to a model of $\bar{q}$ by putting

$$Ms_{i_1 \dots i_k} = (Mf) [My_1^{i_1} \cap \dots \cap My_k^{i_k}] .$$

Conversely, let M be a model for $\bar{q}$ and let f be any map variable in q . Let Mf be any map whose domain and range are Mx_{Df} and Mx_{Rf} respectively, and such that

$$(Mf) [My_1^{i_1} \cap \dots \cap My_k^{i_k}] = Ms_{i_1 i_2 \dots i_k} \quad \text{for each } i_1, \dots, i_k.$$

The existence of such a map Mf follows from conditions (1) and (3) by observing that Mx_{Df} is the disjoint union of all sets $My_1^{i_1} \cap \dots \cap My_k^{i_k}$. This choice of Mf plainly satisfies all literals of type (D,R) in q . Moreover, from condition (2) and by noting that My_j^+ is the disjoint union of all sets $My_1^{i_1} \cap \dots \cap My_k^{i_k}$ with $i_j = +$, it follows that literals of type ([]) in q are correctly modeled. This completes the proof of the theorem. Q.E.D.

The decidability of this extension follows from this theorem, using the decidability of multilevel syllogistic.

Remarks. (a) We can extend our language by adding the inverse operator f^{-1} on maps and also by allowing literals of the form $y = f^{-1}[x]$ to appear in q . To do this, we have only to note the following equivalence:

$$y = f^{-1}[x] \leftrightarrow y \subseteq Df \ \& \ f[y] = x \cap Rf \ \& \ f[Df - y] \cap x = \emptyset .$$

(b) We can add the predicate `singlevalued( )` also allowing literals of type

$$\text{singlevalued}(f), \quad \neg \text{singlevalued}(f)$$

to appear in q . If for some f in q both literals `singlevalued(f)` and `¬singlevalued(f)` are in q , then q is clearly unsatisfiable. If at most one of those literals is in q for each map variable f , and if for each literal `¬singlevalued(f)` we add the formula

$$x_{Df} \neq \emptyset$$

to $\bar{q}$, then Theorem 2 still holds. Indeed, let M be a model for $\bar{q}$ and let $x_1, x_2, \dots, x_h$ be all variables in $\bar{q}$. Then, without loss of

generality we can assume that all nonempty disjoint parts of the Venn diagram determined by the sets

$$Mx_1, Mx_2, \dots, Mx_h,$$

are countably infinite (just add arbitrary new individuals to these disjoint parts). Next, take Mf , as in the proof of Theorem 2, to be either singlevalued or not singlevalued according to the corresponding literal in q (arbitrarily if there is no such literal).

(c) The predicate one-one() can be treated in much the same way. For each literal $\neg \text{one-one}(f)$ add $x_{Df} \neq \emptyset$, and for each literal one-one(f) add the set of formulae

$$(4) \quad s_{i_1 \dots i_k} \cap s_{j_1 \dots j_k} = \emptyset \quad \text{for every } (i_1, \dots, i_k) \neq (j_1, \dots, j_k).$$

(d) Finally we observe that the multilevel syllogistic described in Section 2, and extended to include all the constructs described in Sections 4 and 5 remains decidable because Theorem 2 can be extended to this case. If the predicate singlevalued() is also introduced then each literal $\neg \text{singlevalued}(f)$ is replaced by the set of formulae

$$z \in x_{Df} \ \& \ w = \{z\} \ \& \ w' = f[w] \ \& \ \#w' > 1,$$

where z , w and w' are new variables, whereas if singlevalued(f) is in q then condition (1) must be replaced by

$$(1') \quad \#(y_1^{i_1} \cap y_2^{i_2} \cap \dots \cap y_k^{i_k}) \geq \#s_{i_1 i_2 \dots i_k} \ \& \ (s_{i_1 i_2 \dots i_k} = \emptyset \rightarrow y_1^{i_1} \cap y_2^{i_2} \cap \dots \cap y_k^{i_k} = \emptyset).$$

Similarly, $\neg \text{one-one}(f)$ is replaced by the set of formulae

$$\neg \text{singlevalued}(f) \vee (z \subseteq x_{Df} \ \& \ \#z > 1 \ \& \ w = f[z] \ \& \ \#w = 1),$$

where w , z are new variables, whereas if one-one(f) is in q then we introduce condition (4) of Remark (c) and replace (1) by:

$$(1'') \quad \#(y_1^{i_1} \cap y_2^{i_2} \cap \dots \cap y_k^{i_k}) = \#s_{i_1 i_2 \dots i_k}.$$

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